Pranav Kandpal

+91 8375858199 | pranav.kandpal05@gmail.com | linkedin.com/in/pranav-kandpal | github.com/PranavKndpl

Education

Dronacharya College of Engineering — Gurugram University Gurgaon, India
*Bachelor of Technology in Computer Science and Engineering — **CGPA: 8.2 / 10*** *2023 – 2027*

Kamal Model School — CBSE Gurgaon, India
*Grade 12 (Senior Secondary): **93%** | Grade 10 (Secondary): **77%*** *2021 – 2023*

Experience

Doctor's EHR LLC Remote
Software Development Intern *June 2025 – July 2025*

- Developed RESTful API endpoints for the Patient Management module, integrating **MinIO (S3-compatible object storage)** to enable secure, scalable document uploads for patient records.
- Engineered a robust error-handling framework using `@ControllerAdvice` and custom exception handlers to ensure standardized JSON error responses across the application.
- Implemented **JWT-based authentication** and server-side input validation to secure patient data and prevent malformed requests.
- Designed and executed structured test scenarios for 6+ core modules, documenting API behavior via **Postman Collections** to streamline frontend integration.

Projects

Lethe | *Rust, FUSE, Systems Programming*

- Built a secure userspace filesystem (FUSE) in Rust, decoupling storage logic from kernel operations.
- Implemented "Zero-Knowledge" encryption (XChaCha20-Poly1305) ensuring data is opaque before writing to disk.
- Designed a sharding engine that splits files into 64KB blocks to hide file metadata and size analysis.
- Optimized write performance using asynchronous flushing and in-memory caching to reduce system call overhead.

FluxDB | *C++17, TCP Networking*

- Developed a multi-threaded NoSQL database from scratch, writing a custom binary TCP protocol for low latency.
- Created an indexing system that switches between Hash Maps and Sorted Arrays based on query patterns.
- Implemented crash recovery using Write-Ahead Logging (WAL) to ensure data durability.
- Handled raw memory allocation in C++ to prevent leaks while managing concurrent client connections.

FluxCRM | *C++, TCP Networking, Event-Driven Architecture*

- Built a headless client to interface with a distributed NoSQL database using a custom lightweight text protocol over raw TCP sockets.
- Engineered a non-blocking background event thread to maintain a persistent connection and stream real-time Pub/Sub events.
- Implemented manual JSON serialization and deserialization from scratch, bypassing heavy external frameworks to minimize latency.

Technical Skills

Languages: Rust, C++17, Python, Java, SQL
Core Concepts: Distributed Systems, Multi-threading, Concurrency, Event-Driven Architecture, Memory Management
Systems & Networking: TCP/UDP Sockets, RESTful APIs, FUSE, POSIX, Linux (Bash)
Tools: Docker, Git, Postman, GDB, Valgrind